

Numerical Convergence of the Cohesive Element Approach in Dynamic Fragmentation Simulations

R. Raghupathy, G. A. Gazonas, J. F. Molinari, and F. Zhou

Citation: [AIP Conference Proceedings](#) **845**, 654 (2006); doi: 10.1063/1.2263407

View online: <http://dx.doi.org/10.1063/1.2263407>

View Table of Contents: <http://scitation.aip.org/content/aip/proceeding/aipcp/845?ver=pdfcov>

Published by the [AIP Publishing](#)

Articles you may be interested in

[2.5D FINITE/INFINITE ELEMENT APPROACH FOR SIMULATING TRAIN-INDUCED GROUND VIBRATIONS](#)

AIP Conf. Proc. **1233**, 5 (2010); 10.1063/1.3452242

[Numerical simulation of burst defects in cold extrusion process](#)

AIP Conf. Proc. **908**, 1395 (2007); 10.1063/1.2741004

[Numerical Modeling of Shock-Induced Damage for Granite under Dynamic Loading](#)

AIP Conf. Proc. **845**, 1431 (2006); 10.1063/1.2263593

[Experimental Study and FEM Analysis of Dynamic Response in Metals during Plate Impact](#)

AIP Conf. Proc. **712**, 498 (2004); 10.1063/1.1766574

[Modelling of micromechanical fracture using a cohesive finite element method](#)

AIP Conf. Proc. **505**, 623 (2000); 10.1063/1.1303551

NUMERICAL CONVERGENCE OF THE COHESIVE ELEMENT APPROACH IN DYNAMIC FRAGMENTATION SIMULATIONS

R. Raghupathy¹, G.A. Gazonas², J.F. Molinari^{1*}, F. Zhou¹

¹ *Department of Mechanical Engineering, Johns Hopkins University, Baltimore, MD 21218*

² *US Army Research Laboratory, Weapons and Materials Research Directorate, Aberdeen Proving Ground, MD 21005*

Abstract. The cohesive element approach is getting increasingly popular for simulations in which a large amount of cracking occurs. Naturally, a robust representation of fragmentation mechanics is contingent to an accurate description of dissipative mechanisms in form of cracking and branching. This paper addresses the issue of energy convergence of the finite-element solution for high-loading rate fragmentation problems. These results provide new insight for choosing mesh sizes and size distributions in two and three-dimensional fragmentation.

Keywords: dynamic crack propagation; crack branching; fragmentation; cohesive element; numerical convergence.

PACS: 46.15.-x, 46.50.+a, 62.20.-x

INTRODUCTION

In ceramic materials, it is known that strength, crack initiation and propagation are affected by the presence of flaws at the micrometer scale. Under dynamic loading conditions, cracks initiate at these flaws, and potentially propagate catastrophically to cause large-scale structural failure. Multiple crack initiate at seemingly random locations and material failure occurs through a complex * communication process of stress-wave interactions between cracks. One of the recent numerical approaches for fragmentation that is easy to implement is that of cohesive element method. Crack branching and fragmentation are natural outcomes of the method. In the creation of new surfaces, each opening cohesive element dissipates a given amount of energy. The total energy dissipated is clearly related to the crack path and mesh convergence. It has been generally observed that for a fixed strain

rate, finer meshes lead to higher energy dissipation due to more microcracking [1]. The issue of energy convergence during the cracking process is of fundamental importance as it clearly affects the physical robustness of the numerical methodology, and is the main purpose of the paper.

We briefly describe the finite element methodology including the cohesive element approach (illustrated by 2D crack-branching simulation) and analyze convergence issues in a simplified one-dimensional setting and the implications for extension to multidimensional finite element simulations.

COHESIVE ELEMENT METHODOLOGY

Our adopted framework is the explicit dynamics finite element analysis. The equations of equilibrium are integrated along the time axis by the second-order accurate explicit (Newmark) scheme. For stability of the time integration, the time step Δt must be lower than a critical value,

* corresponding author: molinari@jhu.edu

Δt_{stable} , which is related to the dilatational (the fastest) wave speed and the (smallest) mesh size. We use the initially rigid cohesive law [2] and cohesive elements are added on the fly (dynamic insertion). The initial finite-element mesh is free of cohesive elements, and as the dynamic simulation proceeds, cohesive elements are inserted at locations where the stress exceeds a critical value σ_c .

In addition to σ_c (maximum cohesive force) and δ_c (critical opening displacement), a third parameter G_c , which is not independent of the other two corresponds to the area under the $\sigma_c - \delta_c$ curve and is the fracture energy that is needed to fully open a unit area of crack surface. Resolution of appropriate length and time scales associated with the fragmentation process requires proper choice of the material parameters. Expressions for cohesive zone length α have been obtained by [4,5]. The finite element mesh size must be smaller than α in order to resolve the physics of dynamically failing material near a crack tip. It has also been shown [3] that cohesive elements introduce a time scale into the problem associated with the opening of an isolated micro crack which determines whether a particular externally applied strain rate is perceived as a low strain rate or a high strain rate during the dynamic decohesion process. In order to properly resolve the unloading part of the cohesive law in a dynamic fragmentation simulation, the integration time step must be roughly an order of magnitude smaller than this time scale.

Previously, we obtained probing crack branching results in 3D PMMA plates [6]. For the approach in 2D, similar to Falk *et al* [7] we meshed a $h=3\text{mm}$ square block of PMMA with an edge crack of 0.25 mm. Finite-element meshes of different sizes (smaller than the cohesive zone length estimate in [5]) were used.

Crack branching results were obtained for all meshes at sufficiently high energy levels for different loading conditions, indicating the method's robustness. However an increase in total cohesive energy dissipated with mesh refinement, was observed. Closer examination of the cracked surfaces showed increased microcracking and crack path variation with mesh refinement (also observed by Ruiz *et al* [1]) leading us to question

the possibility of energy convergence, and mesh-independent microcracking. These questions are addressed in the next section for an idealized 1D setting.

DYNAMIC FRAGMENTATION MODEL

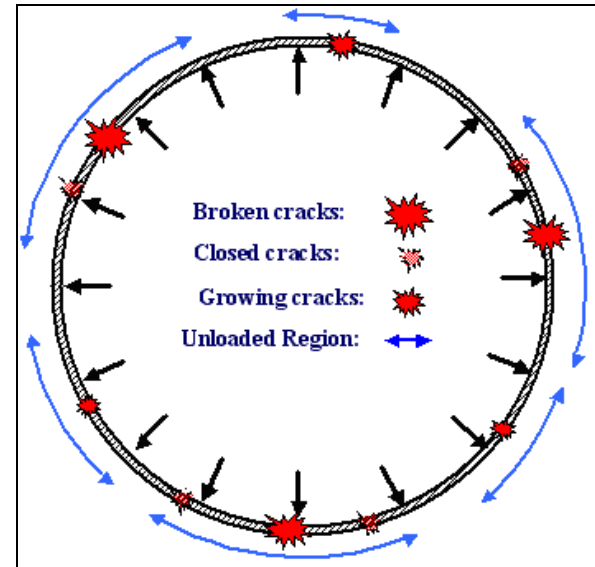


Figure 1: Schematic of dynamic ring fragmenting under explosive loading

For simplicity we consider the most elementary model of an expanding ring (Fig. 1) which has been discussed in earlier work [3,8]. We assume a linearly elastic material and insert 1D initially-rigid cohesive elements upon reaching a critical stress to monitor multiple cracks nucleation, growth, and interactions. The 1D setting allows the use of the method of characteristics to track stress wave interactions in the elastic material and permits high mesh densities at an affordable cost. The results were also compared to finite-element calculations and no significant differences were observed. The ring is loaded explosively with different radial speeds. Fragment statistics are collected when all the fragments are formed. Fragmentation simulations were performed for different mesh sizes (from 128 to 1 million nodes) and for strain rates ranging from $5 \times 10^3 \text{ s}^{-1}$ to $5 \times 10^5 \text{ s}^{-1}$.

Figure 2 represents the energy dissipated (curves in solid lines) in the process of cohesive fracture. This includes cohesive elements that have

been either fully or partially damaged. We observe a non-monotonic convergence.

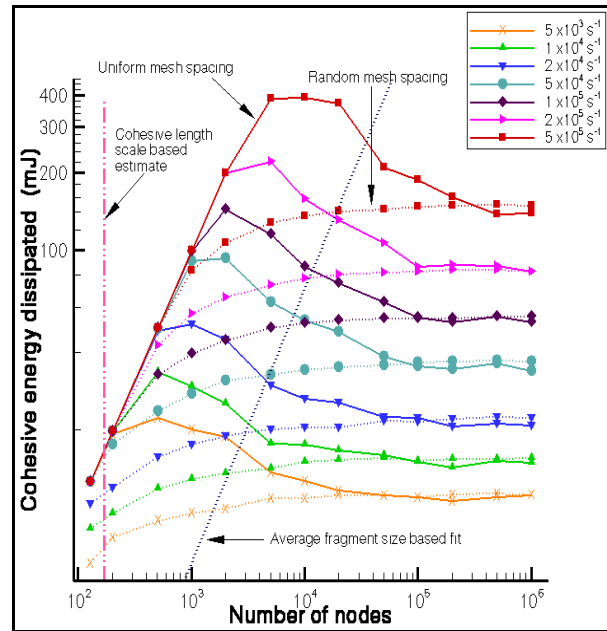


Figure 2: Cohesive energy dependence on mesh size for uniform and random mesh spacing

For coarse meshes the energy dissipated is proportional to the number of nodes in the mesh. This strong mesh dependence clearly indicates that these mesh sizes under resolve the fragmentation process (all the cohesive elements are fully broken and thus the fragment size is mesh size dependent).

The cohesive zone size estimate is shown by the vertical line in Figure 2. Clearly, for this highly dynamic problem, this estimate of mesh size is not satisfying, as it cannot resolve the fragmentation process. In light of this, one may question the validity of using cohesive zone length scale estimates in any dynamic finite-element problem. For illustration, consider the highest strain rate case ($5 \times 10^5 \text{ s}^{-1}$). We note that the dissipated energy increases up to 10^4 nodes. Although 10^4 nodes is a relatively small mesh in a 1D setting, its 2D equivalent is 10^8 nodes. Many published results using the cohesive element approach in a multidimensional space have been obtained with smaller meshes. We postulate that observations of energy increase with mesh refinement are due to microcracking being not properly resolved, although the increase may not be linearly

proportional to the node number as boundary conditions may deviate from the simple uniaxial state of stress here.

A significant finding of this paper is that for sufficiently large meshes one can properly resolve microcracking. Smooth convergence is observed beyond the peak of the curve. This is encouraging as it indicates that although cohesive elements are inserted at many more nodes than necessary most are not severely damaged and do not impact significantly the energy balance of the problem. However it is disturbing to note that this translates to requiring very fine meshes (10^{12} nodes in 2D) beyond any available computational power.

Using a smaller time step did not affect the non-monotonic nature of the curve indicating that perhaps it was not a numerical issue. It was shown previously [8] that a simple universal law (Weibull distribution) captures fragment sizes for all strain rates. Nature wishes to maximize entropy and stays away from uniform fragment sizes due to the inherent randomness associated with generating fragments (which uniform meshes do not capture well). Uniform meshes severely constrain the fragmentation event and its desire to maximize entropy and seems to explain the observed non-monotonic convergence.

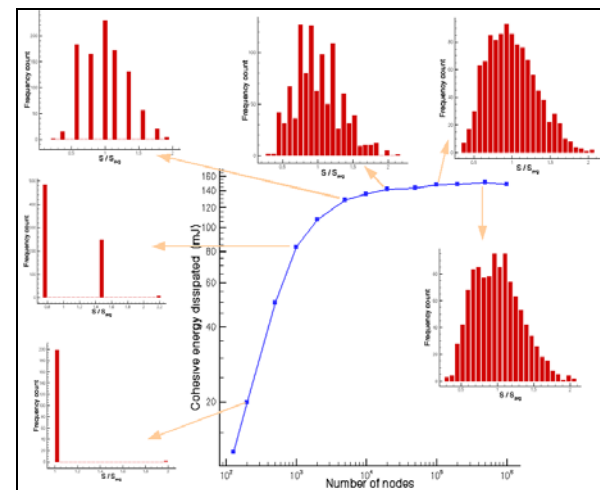


Figure 3: Mesh dependence of fragment size distribution (random mesh spacing) for $5 \times 10^5 \text{ s}^{-1}$

The curve in dotted lines in Figure 2 represent the results obtained for random meshes obtained by shifting each node by a random amount in between

$\pm 0.4h_e$, where h_e is the mesh size in the corresponding uniform mesh. The convergence is now monotonic and, remarkably, it is up to two orders of magnitudes faster. In addition it yields smooth and detailed statistics on fragment sizes (Fig. 3). The required number of nodes and their equivalent in higher dimensions is manageable with parallel simulations. Numerical modeling of fragmentation thus has an odd characteristic: uniform meshes should be avoided at all cost.

Instead of shifting by $\pm 0.4h_e$ we tested for varying degrees of mesh heterogeneity and found that random displacements of nodes by an amount in between $\pm 0.05h_e$ suffice to improve convergence by roughly two orders of magnitude.

As discussed earlier, at least in a 1D setup, cohesive zone estimates such as in [5], lead to meshes too coarse to capture all microcracking events. A key finding of our previous work was that a universal law may be used to capture the dependence of the average fragment size with strain rate. This empirical estimate was proposed as an alternative to Grady's energy-based model [9], which does not take into account wave reflections. Using the average fragment size relation in [10] and Figure 2, we obtain an empirical estimate for required mesh size (shown as dotted line in Figure 2) as:

$$h_e = 0.25(EG_c / \sigma_c^2) \left[1 + 4.5 \left(\frac{\dot{\epsilon}}{c\sigma_c^3 / E^2 G_c} \right)^{\frac{2}{3}} \right]^{-1}$$

Using this estimate and perhaps others incorporating the time scale, converged 2D fragmentation simulations are within reach. 3D simulations, however, remain too computationally intensive. Rate dependent cohesive laws, for which G_c becomes an increasing function of strain rate, are promising in this regard [11].

CONCLUSIONS

Standard cohesive zone size estimates under resolve the number of nodes necessary for attaining mesh independence in dynamic fragmentation simulations. Substantial computational evidence demonstrates, for the first time, that the cohesive

element method converges in an energetic sense. Microcracking events and the ensuing fragment sizes distributions are statistically mesh independent for sufficiently fine meshes. Random mesh spacing led to much faster and smoother convergence than uniform meshes, and that even for very small random perturbations. A simple mesh density estimate based on the dependence of the average fragment size on the strain rate and elastic and fracture material properties is proposed. This estimate provides a clear roadmap for more complex loading conditions including biaxial loading of ceramic plates.

REFERENCES

1. Ruiz, G., Ortiz, M., and Pandolfi, A., *Int. J. Numer. Meth. Engng*, 52, 97-120, 2001
2. Camacho, G. T., and Ortiz, M., *Int. J. Solids Structures*, 33, 2899-2938, 1996
3. Zhou, F., Molinari, J. F., Ramesh, K. T., "Effects of Material Properties and Strain Rate on the Fragmentation of Brittle Materials", *Int. J. Fracture*, (submitted) 2005
4. Rice, J. R., "Fracture, An Advanced Treatise, Vol. II: Mathematical Fundamentals", Liebowitz, H., (Ed.), Academic Press, New York, 191-311, 1968
5. Palmer, A. C., and Rice, J. R., *Proc. Roy. Soc. London Ser. A*, Vol. 332, 527-548, 1973
6. Zhou, F., and Molinari, J. F., *Int. J. Numer. Meth. Engng*, 59(1), 2004
7. Falk, M. L., Needleman, A., and Rice, J. R., *Journal de Physique IV, Proceedings*, pp. Pr.5.43 - Pr.5.50, 2001
8. Zhou, F., Molinari, J. F., Ramesh, K. T., "A finite difference analysis of the brittle fragmentation of an expanding ring", *Computational Materials Science*, (accepted) 2005
9. Grady, D. E., *J. Appl. Phys.*, 53(1), 1982
10. Zhou, F., Molinari, J. F., Ramesh, K. T., "Characteristic fragment size distribution of dynamically expanding rings", *Physical Review Letters*, (submitted) 2005
11. Zhou, F., Molinari, J. F., Shioya, T., *Engineering Fracture Mechanics*, 72, pp. 1383-1410, 2005